

Supplementary Material

Moment-to-Moment Affect During Film-Induced Somatic Anxiety

S0. The Rating Task

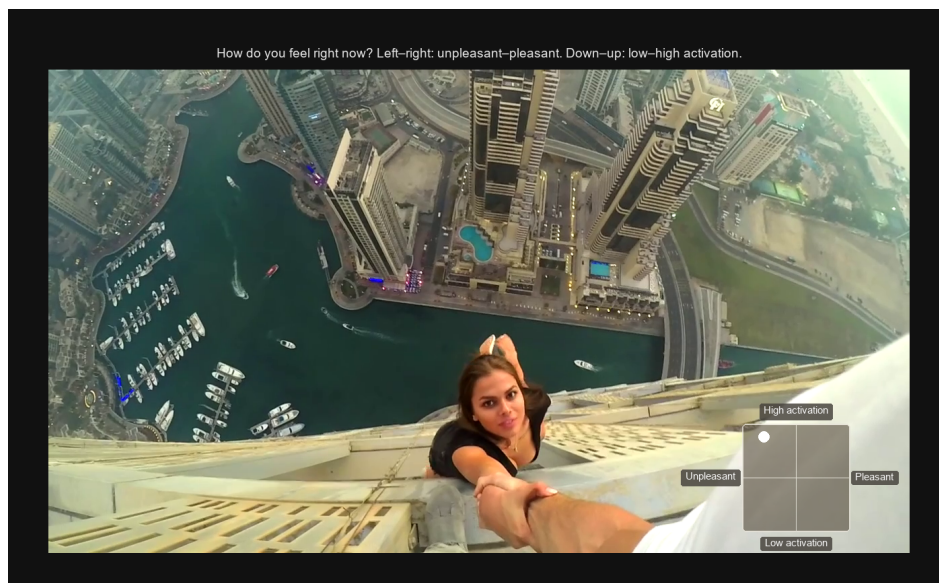


Figure S1: The rating screen during the film (illustration). Participants moved the dot within the affect grid in the lower right corner with the mouse: left to right for unpleasant to pleasant, down to up for low to high activation (arousal). The image is a composite of the film frame at 10:36, during the scene in which a member of the group is held by one hand over the edge, with the rating grid redrawn as it appears in the task; the dot marks the confirmatory sample's mean rating at that moment (valence 20, arousal 88). Film frame from Tikhomirov (2017).

S1. Sample and Descriptive Statistics

Table S1 describes the exploratory sample, all confirmatory completers and the primary confirmatory sample; Table S2 gives descriptive statistics for all measures used in the confirmatory tests.

S2. Confirmatory Tests and Sensitivity Analyses

Table S3 lists all preregistered tests in the primary sample, and Table S4 repeats them in the two preregistered sensitivity samples. The stricter-engagement sample ($n = 96$) additionally excluded sparse raters (fewer than one rating bout per minute, or more than 180 s without input; 24), participants who failed one attention check (5) and participants who reported not watching attentively (3); these groups overlap. The second sample ($n = 120$) excluded the 4 participants whose Prolific submissions were rejected for behaviour on the free-text page but who passed the registered exclusion rules. Figure S2 shows segment means and Figure S3 the individual-difference tests.

Table S1: Sample Characteristics

Characteristic	Exploratory, usable	Confirmatory, all completers	Confirmatory, primary sample
<i>n</i>	21	130	124
Age, years, <i>M</i> (<i>SD</i>)	35.0 (12.6)	34.1 (11.9)	34.2 (12.0)
Range	19–60	18–76	18–76
Sex, <i>n</i> (%)			
Female	13 (62)	62 (48)	59 (48)
Male	8 (38)	64 (49)	62 (50)
Not available ^a	0 (0)	4 (3)	3 (2)
Input device, <i>n</i> (%)			
Mouse	15 (71)	86 (66)	80 (65)
Trackpad	6 (29)	43 (33)	43 (35)
Other	0 (0)	1 (1)	1 (1)
Everyday fear of heights (1–7), <i>M</i> (<i>SD</i>)	3.52 (1.72)	3.72 (1.73)	3.77 (1.73)
STICSA somatic, pre-film (11–44), <i>M</i> (<i>SD</i>)	12.24 (1.81)	13.72 (3.51)	13.76 (3.56)
Failed one attention check, <i>n</i> (%)	0 (0)	5 (4)	5 (4)
Excluded under registered rules, <i>n</i>	0	6	0

Note. Exploratory sample: pilot and cohort 1, usable under the registered rules (reference curve for H3). Confirmatory sample: cohorts 2–4. Age and sex are Prolific profile data; input device is self-reported. STICSA = State-Trait Inventory for Cognitive and Somatic Anxiety, state somatic subscale. ^aProlific withholds demographic data for rejected submissions.

Table S2: Descriptive Statistics, Primary Confirmatory Sample

Measure	Scale	<i>M</i>	<i>SD</i>	<i>Mdn</i>	Range
<i>Before the film</i>					
Everyday fear of heights	1–7	3.77	1.73	4.00	1–7
STICSA somatic, pre-film	11–44	13.76	3.56	13.00	11–27
<i>After the film</i>					
STICSA somatic, post-film	11–44	20.55	7.29	18.00	11–44
STICSA somatic change (post – pre)		6.79	6.60	5.00	–6–32
Overall anxiety while watching	1–7	5.18	1.33	5.00	1–7
Concern for the climbers' safety	1–7	6.13	1.30	7.00	1–7
Intensity of bodily sensations	1–7	4.64	1.60	5.00	1–7
<i>Continuous arousal</i>					
Baseline (0:00–4:22)	0–100	43.13	15.80	43.76	5.9–76.4
Rise (4:22–7:50)	0–100	67.14	14.79	68.91	22.5–96.1
Plateau (7:50–11:32)	0–100	80.91	15.01	83.07	24.2–100.0
Return (11:32–13:28)	0–100	53.89	19.42	54.76	13.3–96.9
Whole film	0–100	61.23	10.99	60.58	21.2–86.7
<i>Continuous valence</i>					
Baseline (0:00–4:22)	0–100	50.58	14.55	51.96	12.6–85.2
Rise (4:22–7:50)	0–100	39.58	19.06	38.52	0.2–83.3
Plateau (7:50–11:32)	0–100	29.13	23.10	23.89	0.0–100.0
Return (11:32–13:28)	0–100	50.84	20.44	51.67	0.6–97.9
Whole film	0–100	41.89	14.67	41.91	4.6–86.5

Note. *N* = 124. Range is the observed minimum–maximum. Continuous ratings are participant means of the 1-Hz series within each registered segment (film time in min:s); valence 0 = unpleasant, 50 = neutral, 100 = pleasant; arousal 0 = low, 100 = high. STICSA = State-Trait Inventory for Cognitive and Somatic Anxiety, state somatic subscale.

Table S3: Preregistered Confirmatory Tests, Primary Sample

Hypothesis	Prediction	Test statistic	Effect size [95% CI]	p	p_{Holm}	Supported
H1	STICSA somatic: post > pre	$t(123) = 11.46$	$d_z = 1.03 [0.81, 1.25]$	$< .001$	–	Yes
H2a	Arousal: rise > baseline	$t(123) = 14.67$	$d_z = 1.32 [1.07, 1.56]$	$< .001$	$< .001$	Yes
H2b	Arousal: plateau > baseline	$t(123) = 19.77$	$d_z = 1.78 [1.49, 2.06]$	$< .001$	$< .001$	Yes
H2c	Arousal: return < plateau	$t(123) = -14.70$	$d_z = -1.32 [-1.56, -1.08]$	$< .001$	$< .001$	Yes
H2d	Valence: plateau < baseline	$t(123) = -9.85$	$d_z = -0.88 [-1.09, -0.68]$	$< .001$	$< .001$	Yes
H3 arousal	Arousal curve replicates	–	$r = .993 [.990, .996]$	–	–	Yes
H3 valence	Valence curve replicates	–	$r = .937 [.893, .966]$	–	–	Yes
H4a	Fear of heights, plateau arousal (+)	–	$\rho = .08 [-.09, .26]$	.355	.355	No
H4b	Fear of heights, plateau valence (–)	–	$\rho = -.32 [-.46, -.16]$	$< .001$	.001	Yes
H4c	Fear of heights, STICSA change (+)	–	$\rho = .18 [.00, .35]$	.049	.098	No
H5a	STICSA change, whole-film arousal (+)	–	$\rho = .24 [.05, .41]$	.008	.008	Yes
H5b	STICSA change, whole-film valence (–)	–	$\rho = -.38 [-.51, -.23]$	$< .001$	$< .001$	Yes
H6a	Overall anxiety, STICSA change (+)	–	$\rho = .56 [.44, .67]$	$< .001$	$< .001$	Yes
H6b	Overall anxiety, plateau arousal (+)	–	$\rho = .34 [.15, .50]$	$< .001$	$< .001$	Yes

Note. $N = 124$. H1 and H2: paired t tests with d_z and exact 95% CIs (noncentral t). H3: Pearson r between the confirmatory and exploratory group-mean series (808 one-second values), 95% CI from a circular block bootstrap (30-s blocks, 5,000 resamples); supported if the lower bound exceeds .50. H4–H6: Spearman ρ with percentile bootstrap 95% CIs (5,000 resamples). Holm correction within H2 (four tests), H4 (three), H5 (two) and H6 (two); H1 is a single test. All tests two-sided; a directional hypothesis counts as supported only if significant in the predicted direction.

Table S4: Preregistered Tests Repeated in the Sensitivity Samples

Hypothesis	Effect	Primary ($n = 124$)	Stricter engagement ($n = 96$)	Rejection set removed ($n = 120$)
H1	d_z	1.03 ($p < .001$)	1.09 ($p < .001$)	1.02 ($p < .001$)
H2a	d_z	1.32 ($p_{\text{Holm}} < .001$)	1.51 ($p_{\text{Holm}} < .001$)	1.43 ($p_{\text{Holm}} < .001$)
H2b	d_z	1.78 ($p_{\text{Holm}} < .001$)	1.94 ($p_{\text{Holm}} < .001$)	1.85 ($p_{\text{Holm}} < .001$)
H2c	d_z	-1.32 ($p_{\text{Holm}} < .001$)	-1.65 ($p_{\text{Holm}} < .001$)	-1.39 ($p_{\text{Holm}} < .001$)
H2d	d_z	-0.88 ($p_{\text{Holm}} < .001$)	-0.88 ($p_{\text{Holm}} < .001$)	-0.87 ($p_{\text{Holm}} < .001$)
H3 arousal	r	.993 (lower .990)	.990 (lower .984)	.993 (lower .989)
H3 valence	r	.937 (lower .893)	.925 (lower .869)	.927 (lower .876)
H4a	ρ	.08 ($p_{\text{Holm}} = .355$) [†]	.03 ($p_{\text{Holm}} = .775$) [†]	.08 ($p_{\text{Holm}} = .380$) [†]
H4b	ρ	-.32 ($p_{\text{Holm}} = .001$)	-.30 ($p_{\text{Holm}} = .010$)	-.32 ($p_{\text{Holm}} = .001$)
H4c	ρ	.18 ($p_{\text{Holm}} = .098$) [†]	.21 ($p_{\text{Holm}} = .088$) [†]	.17 ($p_{\text{Holm}} = .142$) [†]
H5a	ρ	.24 ($p_{\text{Holm}} = .008$)	.24 ($p_{\text{Holm}} = .018$)	.24 ($p_{\text{Holm}} = .008$)
H5b	ρ	-.38 ($p_{\text{Holm}} < .001$)	-.32 ($p_{\text{Holm}} = .003$)	-.39 ($p_{\text{Holm}} < .001$)
H6a	ρ	.56 ($p_{\text{Holm}} < .001$)	.49 ($p_{\text{Holm}} < .001$)	.56 ($p_{\text{Holm}} < .001$)
H6b	ρ	.34 ($p_{\text{Holm}} < .001$)	.29 ($p_{\text{Holm}} = .004$)	.36 ($p_{\text{Holm}} < .001$)

Note. Cells show the effect size and, in parentheses, the Holm-adjusted p value (H1: unadjusted, single test) or, for H3, the lower bound of the 95% CI of r . [†]Hypothesis not supported under the registered criteria. Stricter engagement: sparse raters (fewer than one rating bout per minute or more than 180 s without input), participants who failed one attention check and participants who reported not watching attentively removed. Rejection set removed: participants whose Prolific submissions were rejected for behaviour on the free-text page removed, although they pass the registered exclusion rules.

Segment means ($n = 124$): participants (thin lines), mean $\pm$ 95% CI

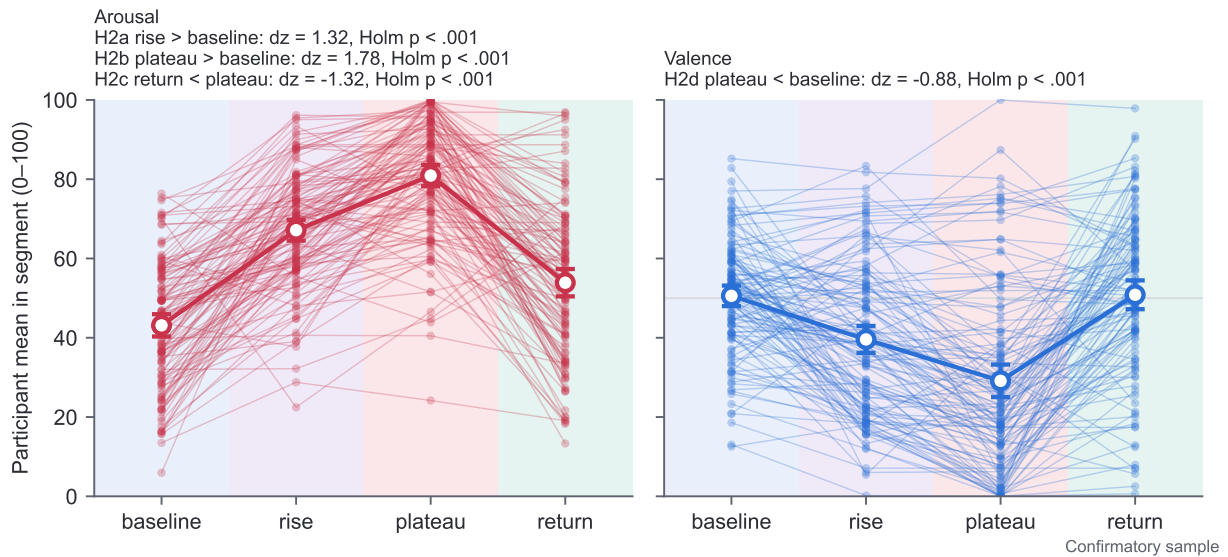


Figure S2: Mean ratings in the four preregistered segments (confirmatory sample, $N = 124$). Thin lines are participants' segment means; the thick line shows group means with 95% confidence intervals.

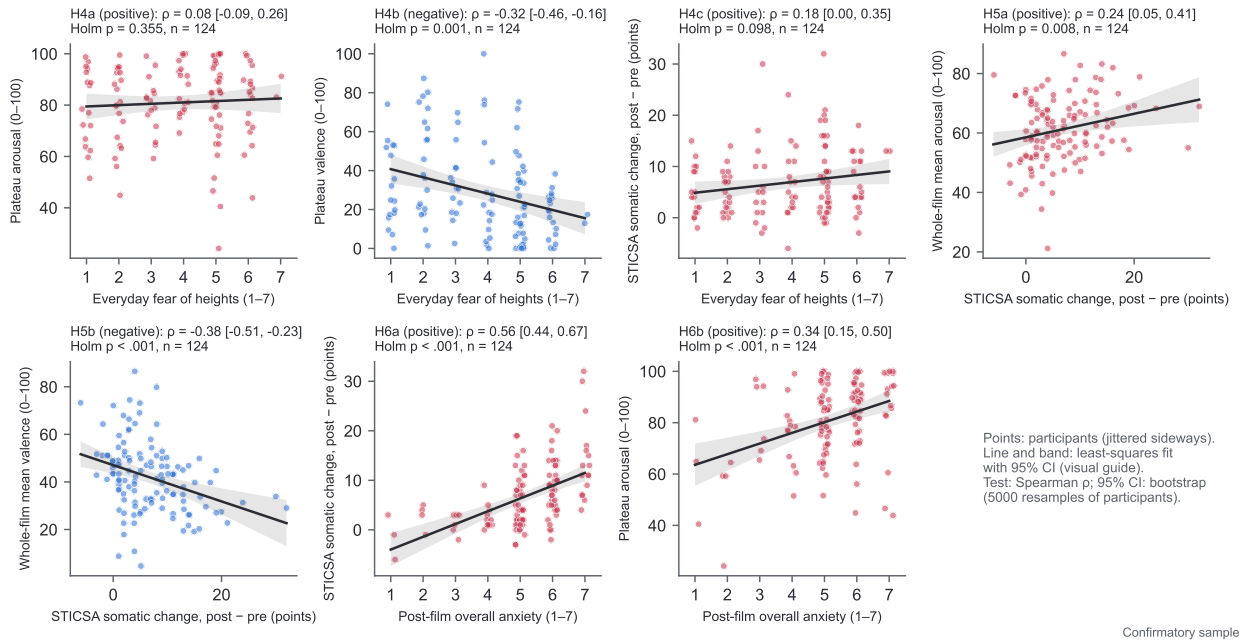


Figure S3: Individual differences tested in H4–H6. Points are participants (jittered sideways); the black line and grey band are a least-squares fit with its 95% confidence interval, shown as a visual guide. Panel headings give Spearman ρ with bootstrap 95% confidence intervals and Holm-adjusted p values.

S3. Registered Descriptive Analyses

Split-half reliability of the group-mean series (1,000 random splits, Spearman–Brown corrected) was .992 for arousal and .979 for valence. Re-estimating the segment edges in the confirmatory sample with the registered piecewise-linear method (three breakpoints, minimum segment length 30 s; 1,000 participant bootstrap resamples) gave 262 s [133, 271], 467 s [317, 475] and 660 s [631, 690], against the registered 262, 470 and 692 s. All tests used the fixed registered edges.

S4. Exploratory Analyses

The registration anticipated exploratory affect-grid and age/sex analyses without specifying confirmatory tests. Figure S4 shows where on the affect grid participants spent their time in each segment. For age and sex (Prolific profile data; $n = 121$ with both available), the STICSA change was regressed on sex and age, and segment means were modelled with a random-intercept mixed model including segment (sum-coded) $\times$ (sex + age), with likelihood-ratio tests of the interactions; the same model fitted by least squares with participant-clustered standard errors is reported as a check (Table S5, Figures S5 and S6). The age/sex models were specified before running them on the confirmatory sample; their p values are uncorrected.

The change in each STICSA somatic item was examined after the confirmatory results were known (requested in review of the draft): for each item, the mean change, the share of participants whose answer increased and d_z , with Wilcoxon signed-rank tests Holm-corrected across the 11 items (Table S6).

Table S5: Exploratory Analyses of Age and Sex

Outcome	Effect	Main analysis		Clustered regression (check)	
		Test or estimate	<i>p</i>	Test or estimate	<i>p</i>
STICSA change	Female – male (points)	0.66 [–1.77, 3.08]	.593	–	–
	Per 10 years of age (points)	0.24 [–0.78, 1.25]	.645	–	–
Arousal	Segment × sex	$\chi^2(3) = 0.28$	.964	$F(3, 120) = 0.08$	.969
	Segment × age	$\chi^2(3) = 4.92$	.178	$F(3, 120) = 2.13$	.100
	Female – male (film mean)	–	–	–0.50 [–4.47, 3.47]	.806
	Per 10 years (film mean)	–	–	–1.26 [–2.88, 0.37]	.129
Valence	Segment × sex	$\chi^2(3) = 0.72$	.869	$F(3, 120) = 0.43$	.734
	Segment × age	$\chi^2(3) = 1.84$	.607	$F(3, 120) = 0.50$	.682
	Female – male (film mean)	–	–	–4.46 [–9.91, 0.99]	.108
	Per 10 years (film mean)	–	–	–0.74 [–2.96, 1.47]	.511

Note. Exploratory analyses, not preregistered tests; *p* values are uncorrected. *n* = 121 participants of the primary sample with age and sex available. STICSA change: linear regression on sex and age. Segment means: random-intercept mixed model with segment (sum-coded) × (sex + age), interactions tested with likelihood-ratio tests (main analysis); the same model fitted by least squares with standard errors clustered by participant, interactions tested with Wald *F* tests, is shown as a check, together with the main effects of sex and age averaged over the film. Age effects are per 10 years; sex effects are female minus male, in scale points.

Table S6: Change in Each STICSA Somatic Item (Exploratory)

Rank	Item	<i>M</i> before	<i>M</i> after	% increased	<i>d_z</i> [95% CI]	<i>p</i> _{Holm}
1	Heart beats fast (item 1)	1.43	2.57	73	1.17 [0.94, 1.40]	< .001
2	Trembly, shaky (item 8)	1.13	1.82	48	0.80 [0.60, 1.00]	< .001
3	Breathing fast, shallow (item 18)	1.17	1.81	50	0.77 [0.57, 0.97]	< .001
4	Muscles tense (item 2)	1.52	2.26	55	0.72 [0.52, 0.91]	< .001
5	Dizzy (item 6)	1.19	1.84	48	0.71 [0.51, 0.91]	< .001
6	Face hot (item 12)	1.15	1.69	40	0.66 [0.47, 0.86]	< .001
7	Butterflies in stomach (item 20)	1.18	1.74	42	0.64 [0.44, 0.83]	< .001
8	Palms clammy (item 21)	1.20	1.73	37	0.60 [0.40, 0.79]	< .001
9	Throat dry (item 15)	1.32	1.73	35	0.54 [0.35, 0.73]	< .001
10	Muscles weak (item 7)	1.22	1.62	35	0.52 [0.33, 0.71]	< .001
11	Arms and legs stiff (item 14)	1.26	1.74	37	0.52 [0.33, 0.71]	< .001

Note. Exploratory analysis, not a preregistered test. *N* = 124 (primary confirmatory sample). Items answered from 1 (not at all) to 4 (very much so); short labels, original STICSA item numbers in parentheses. Ranked by *d_z* (mean change divided by the standard deviation of the changes; exact 95% CI). % increased: share of participants whose answer was higher after the film. *p* from Wilcoxon signed-rank tests, Holm-corrected across the 11 items.

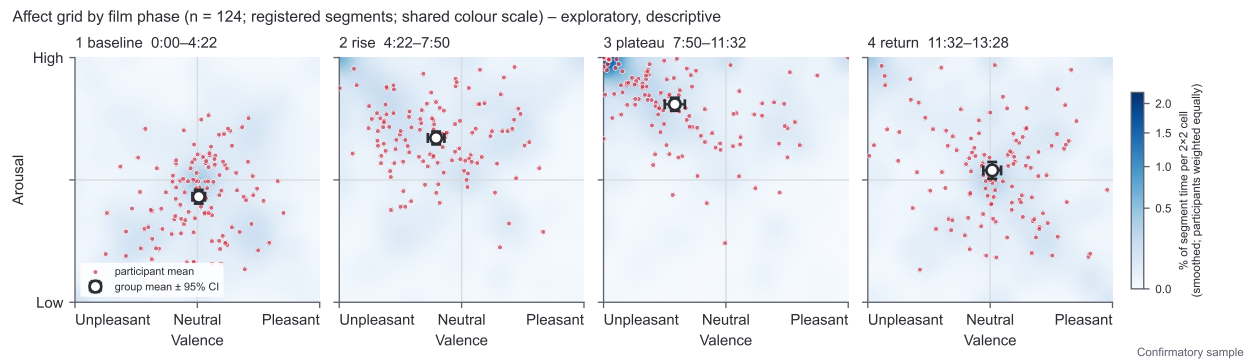


Figure S4: Occupancy of the affect grid in each segment (exploratory). Blue shading shows the share of viewing time in each region (smoothed; participants weighted equally; same scale in all panels); dots are participants' mean ratings; the open circle is the group mean with 95% confidence intervals.



Figure S5: Group-mean ratings by sex (exploratory), with 95% confidence intervals; female $n = 59$ (solid), male $n = 62$ (dashed).

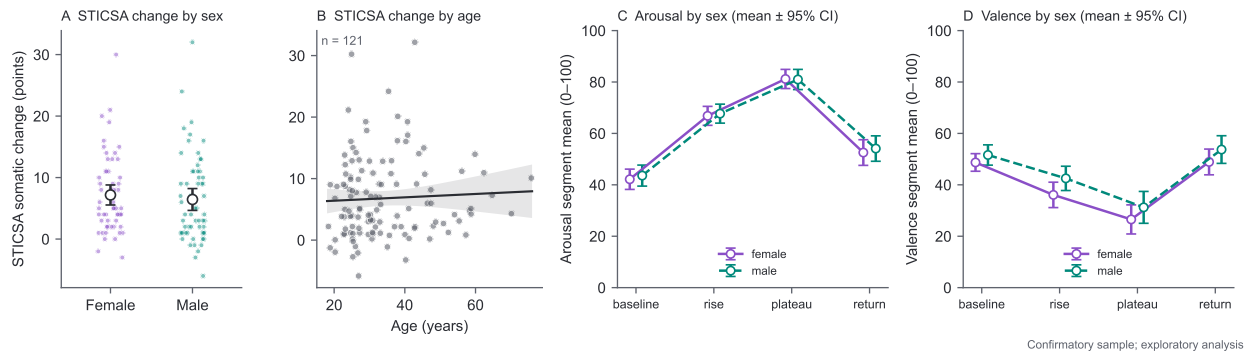


Figure S6: STICSA change by sex (A) and age (B; least-squares line with 95% band as a visual guide), and segment means by sex (C, D), exploratory; $n = 121$.

S5. Notes on the Preregistered Plan

H6 was added when preparing the registration, after inspection of a descriptive group-mean plot and before any confirmatory hypothesis test. H1–H5 and their analysis plan were unchanged.

1. The registration reported 123 usable confirmatory participants at submission; one further participant completed the study after that count, had complete data and passed all exclusion rules, and was included as registered ($N = 124$).
2. The fixed segment edges were estimated on 22 exploratory participants under the quality-control rules in use at the time; under the final registered rules 21 are usable, and these form the reference curve for H3.
3. One participant’s Prolific rejection was reversed for payment reasons; the participant remains in the registered rejection set for the sensitivity analysis.
4. ‘Reported not watching attentively’ was implemented as answering ‘No’; participants who answered ‘Mostly’ were retained.
5. Before the confirmatory run, the segment-edge bootstrap was set to 1,000 participant resamples; the registration did not specify a number. Bootstraps used seed 20261004, and Spearman confidence intervals were percentile intervals. H1 was treated as a single-test family, with the Wilcoxon check reported alongside; H3 was decided by its confidence-interval criterion without a p value.
6. Exploratory age and sex analyses used segment-level mixed models; models over the 1-Hz series, mentioned in the registration as an example, were not implemented. Alternative optimisation algorithms were tried when needed for convergence, without changing the model or decision rule.